

ModelSmith - Functional Requirements

Priority: P0 = MVP/core; P1 = high-value extension.

ID	Module	Functional Requirement	Priority
FR-01	User Authentication	Register, log in, log out and reset password securely.	P0
FR-02	Project Management	Create, view, edit and delete projects; organize associated models and results.	P0
FR-03	Model Upload	Upload supported models, validate files, detect format/framework and store metadata.	P0
FR-04	Model Analysis	Extract graph, layers/operators, parameters, FLOPs, tensor shapes and memory estimates.	P0
FR-05	Profiling	Estimate latency and memory usage and identify performance bottlenecks/hotspots.	P0
FR-06	Deployment Goals	Accept optimization objective, accuracy constraints and target hardware/runtime.	P0
FR-07	Optimization Planning	Generate multiple candidate strategies based on model, goals, hardware and optimization knowledge.	P0
FR-08	Plan Filtering & Ranking	Remove incompatible strategies, rank valid plans and provide reasons for recommendations.	P0
FR-09	Performance Prediction	Predict model size, memory, latency and other relevant outcomes; show expected trade-offs.	P0
FR-10	Plan Comparison	Display candidate plans side-by-side and allow the user to select a preferred plan.	P0
FR-11	Optimization Execution	Convert a selected plan into an executable pipeline and run supported optimization techniques.	P1
FR-12	Benchmarking	Measure and compare original vs optimized latency, throughput, memory, size and accuracy where applicable.	P1
FR-13	Artifacts & Export	Store optimized models, configurations and reports and provide authorized downloads.	P1
FR-14	Reports & Notifications	Generate analysis/optimization/benchmark reports and notify users about long-running job status.	P0/P1
FR-15	Administration	Provide system/user/job monitoring and audit functions where administration is enabled.	P1